INTERNSHIP PROJECT REPORT

Car Price Prediction Using Machine Learning

Submitted By

Name: Ayesha Siddiqua

Course: Data science Internship

Domain: Data Science

Submitted To

Organization: *CODE ALPHA*

Department: Data Science

Project Details

Project Title:
Car Price Prediction Using Machine Learning

Programming Language:
Python

Development Environment:
Visual Studio Code (VS Code)

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Machine Learning Algorithm:
Linear Regression

Dataset:
Car Price Prediction Dataset (301 records)

Date of Submission

Month, Year: July 2026

Declaration

I hereby declare that this internship project titled **"Car Price Prediction Using Machine Learning"** is an original work carried out by me during my DATA SCIENCE Internship. The work presented in this report has not been submitted previously for the award of any degree, diploma, or certification. All sources of information used in this project have been acknowledged appropriately.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Machine Learning is a branch of Artificial Intelligence (AI) that enables computer systems to learn from historical data and make predictions without being explicitly programmed for every task. It has become one of the most important technologies in modern data science and is widely used in finance, healthcare, e-commerce, transportation, and business analytics.

Car price prediction is a practical application of supervised machine learning. The objective is to estimate the selling price of a used car based on various characteristics such as manufacturing year, present showroom price, distance driven, fuel type, transmission type, selling type, and ownership history. Accurate price prediction helps both buyers and sellers make informed decisions while improving transparency in the automobile market.

In this project, a Linear Regression model was developed using Python and Scikit-learn to predict car prices. Before model development, the dataset was explored, cleaned, and preprocessed by converting categorical variables into numerical values. The dataset was then divided into training and testing sets, allowing the model to learn from historical data and evaluate its prediction performance. Finally, graphical visualizations and evaluation metrics were used to assess the effectiveness of the developed model.

CHAPTER 2: OBJECTIVES

The primary objectives of this project are:

- To understand the concept of regression-based machine learning.
- To perform data preprocessing on a real-world dataset.
- To convert categorical variables into numerical values suitable for machine learning algorithms.

- To train a Linear Regression model for predicting car selling prices.
- To evaluate model performance using R^2 Score, MAE, MSE, and RMSE.
- To visualize prediction results using graphs and statistical plots.
- To gain practical experience with Python libraries used in data science.

CHAPTER 3: SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

- Operating System: Windows 10/11
- Programming Language: Python 3.14
- Development Environment: Visual Studio Code
- Libraries Used:
  - Pandas
  - NumPy
  - Matplotlib
  - Scikit-learn

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: Minimum 1 GB free disk space
- Internet Connection: Required for downloading libraries and dataset

---

CHAPTER 4: TOOLS AND TECHNOLOGIES USED

Python

Python is a high-level programming language widely used in machine learning and data science due to its simple syntax, extensive libraries, and strong community support.

Pandas

Pandas is used for loading, cleaning, manipulating, and analyzing structured datasets. It provides DataFrame objects that make handling tabular data efficient.

NumPy

NumPy provides efficient numerical operations, mathematical functions, and array processing capabilities required for scientific computing.

Matplotlib

Matplotlib is used for creating graphs and visualizations such as histograms and scatter plots that help interpret data and model performance.

Scikit-learn

Scikit-learn is a powerful machine learning library containing algorithms for regression, classification, clustering, preprocessing, model evaluation, and dataset splitting.

CHAPTER 5: DATASET DESCRIPTION

The dataset used in this project contains **301 records** and **9 attributes** related to used cars.

The attributes included are:

- **Car_Name:** Name of the car model.
- **Year:** Manufacturing year of the vehicle.
- **Selling_Price:** Selling price of the car (Target Variable).
- **Present_Price:** Current showroom price.
- **Driven_kms:** Total distance driven by the vehicle.
- **Fuel_Type:** Petrol, Diesel, or CNG.
- **Selling_type:** Dealer or Individual.
- **Transmission:** Manual or Automatic.
- **Owner:** Number of previous owners.

The dataset contains **no missing values**, making it suitable for machine learning after preprocessing.

CHAPTER 6: METHODOLOGY

The methodology adopted in this project consists of the following stages:

Step 1: Data Collection

The car price dataset was obtained in CSV format and imported into Python using the Pandas library.

Step 2: Data Exploration

The dataset structure was analyzed using the head(), info(), describe(), and isnull() functions to understand its characteristics and identify missing values.

Step 3: Data Preprocessing

Categorical variables such as Fuel Type, Selling Type, and Transmission were converted into numerical values because machine learning algorithms require numerical input.

The Car_Name column was removed because it contained unique identifiers that were not useful for regression.

Step 4: Feature Selection

Independent variables (features) and the dependent variable (Selling Price) were separated.

Step 5: Data Splitting

The dataset was divided into:

- **80% Training Data**
- **20% Testing Data**

This ensured that the model was evaluated using unseen data.

Step 6: Model Training

A Linear Regression algorithm was trained using the training dataset to establish relationships between input features and the selling price.

Step 7: Prediction

The trained model predicted selling prices for the testing dataset.

Step 8: Model Evaluation

The model performance was evaluated using:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Step 9: Data Visualization

Several graphs were generated to visually interpret the dataset and model performance.

CHAPTER 7: RESULTS AND DISCUSSION

The Linear Regression model was successfully trained using **240 training samples** and evaluated using **61 testing samples**.

The obtained performance metrics are:

- **R^2 Score:** 0.8467
- **Mean Absolute Error (MAE):** 1.2219
- **Mean Squared Error (MSE):** 3.5316
- **Root Mean Squared Error (RMSE):** 1.8792

The R^2 Score of **84.67%** indicates that the developed model explains a significant portion of the variation in car selling prices. The relatively low MAE and RMSE values suggest that the prediction errors are reasonably small, demonstrating good predictive performance.

The scatter plot comparing actual and predicted prices showed that most predictions closely followed the actual values. The histogram illustrated the distribution of selling prices, while the correlation heatmap highlighted relationships among different features. These visualizations provided additional confidence in the effectiveness of the developed regression model.

CHAPTER 8: CONCLUSION

This internship project successfully demonstrated the implementation of a machine learning solution for predicting used car prices using Linear Regression. The dataset was imported, explored, cleaned, and preprocessed before model development. The trained model achieved an **R^2 Score of 84.67%**, indicating strong predictive capability.

The project enhanced practical knowledge of Python programming, data preprocessing, regression analysis, model evaluation, and data visualization. It also provided hands-on experience with widely used data science libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. The knowledge gained from this project forms a strong foundation for future work in **Data Science**, **Machine Learning**, and **Bioinformatics**.

CHAPTER 9: FUTURE SCOPE

The project can be further improved by:

- Training advanced regression models such as Random Forest Regression and XGBoost.
- Increasing the dataset size to improve model generalization.
- Performing feature engineering to extract additional useful information.
- Developing a web application for real-time car price prediction.
- Deploying the model using Flask or Streamlit for public use.

CHAPTER 10: REFERENCES

1. Python Software Foundation. *Python Documentation*. https://docs.python.org/
2. Scikit-learn Developers. *Scikit-learn User Guide*. https://scikit-learn.org/
3. Pandas Development Team. *Pandas Documentation*. https://pandas.pydata.org/docs/
4. NumPy Developers. *NumPy Documentation*. https://numpy.org/doc/
5. Matplotlib Development Team. *Matplotlib Documentation*. https://matplotlib.org/

This level of detail is appropriate for an internship report and should span approximately **25–35 pages** after including the cover page, certificate, declaration, acknowledgement, index, screenshot pages, graphs, and formatting. It is suitable for submission in a Python Programming/Data Science internship.

Include proper screenshot placeholders with captions in the internship report

Absolutely! Since you've completed the project, below is a **professional "Implementation and Results" section** with **proper screenshot placeholders** and **captions**. You can directly copy this into your Word report and insert your screenshots in the indicated places.
